

Arogyam Aradhya — Backend & MLM PRD

Purpose: the single source of truth for the backend + MLM logic. **How to use:** legend below marks what's built vs. what needs your input. Edit anything, add rules under the ? markers, and I'll implement to match.

Legend: ✔ built & live · ● built but simplified/assumed · ? needs your decision · ➡_{SOON} planned/not built

Last updated: 2026-09-08 · Owner: Ankit (cyamsys)

1. Product Overview

Two earning portals on one shared product catalog + one admin panel:

Portal	Who	How they earn
MLM Network	Distributors in a binary tree	Binary matching (weekly), career-rank bonuses, ₹500 direct-referral
Direct Selling (DSA)	Standalone agents (no team)	Flat 40% commission on their own sales

- Pricing base = **DP (distributor price)**. **MRP** = retail (display). **SP (Selling Points)** = business volume for MLM.
- Public storefront (Home, Products, Franchise, Documents, Pay, Gallery, About, Contact) + Cart.

2. Roles & Accounts

- Member** — `segment = "mlm" or "direct"` . Login by member ID / phone / email.
- Admin** — separate `admin_users` table. Single super admin (`admin`).
- Member IDs: `AA` + 8 digits (random). Founder/root is fixed `AA10000001` .
- Portal guard: a `direct` account cannot log into the MLM portal and vice-versa. ✔

3. Tech Stack & Architecture ✔

- Backend:** FastAPI (Python 3.13), SQLAlchemy 2.0, PostgreSQL.
- Frontend:** React (Vite) + Tailwind, served by FastAPI (single service).
- Deploy:** Render (Docker, one web service + managed Postgres). GST/company info + all MLM rates are DB settings.
- Tests:** pytest (31 tests) — auth, matching, ranks, weekly payout, GST, KYC, orders.

4. Data Model (key entities)

Member — `member_id`, `segment`, `name`, `email`, `phone`, `password_hash`, `sponsor_id` (referral), `parent_id` + `position`(L/R) (tree placement), `is_active` (green flag), `is_blocked`, `activated_at`, `rank_level`, `rank_bonus_paid_level`, `left_carry`, `right_carry` (unmatched SP per leg), `total_left_sp`, `total_right_sp` (lifetime, drives rank), `week_left_sp`, `week_right_sp` (reset at weekly close), `self_purchase_sp`, `wallet_balance`, `total_earned`, `total_paid`, `address/city/state/pincode`, `pan`, `aadhaar`, `bank_name/account/ifsc`, `nominee`

Product — `name`, `slug`, `category`, `mrp`, `price`(DP), `cost_price`, `sp`, `stock`, `image`, `is_active`, `is_offer` **Category** — `name`, `slug` **Order / OrderItem** — `order_no`, `member_id`, `subtotal`(taxable), `total`(w/GST), `total_sp`,

status(pending/paid/cancelled), payment_status(unpaid/paid) **SPLedger** — every SP event up a leg (L/R/S)
CommissionLedger — earnings: kind ∈ {matching, rank, referral, dsa, level} **WeeklyPayout** — payout register row:
period, week_label, left_sp, right_sp, matching_sp, closing_sp, payout, cf_left, cf_right, status **PayoutRequest**
— member withdrawal request: amount, status(pending/approved/paid/rejected) **KycDocument** — member_id, doc_type,
filename, content_type, data(blob) (5 slots) **Setting** — key/value config (see §16) **AdminUser** — username,
password_hash

5. Authentication & Security ✓

- Passwords: bcrypt. JWT (HS256), 1-day expiry, SECRET_KEY from env.
 - **Login is case-insensitive for the member ID** — aa10000001, AA10000001, Aa10000001 all log into the same account. Email is matched case-insensitively too; phone is digits. ✓
 - Signup: password ≥ 6 chars, phone normalized to 10 digits (+91/spaces stripped), email format-validated, **phone unique** across both portals.
 - Rate limiting per IP: login 15/5min, signup 10/10min, admin 6/5min.
 - Self-service change password; admin can reset a member's password.
 - Blocked accounts cannot log in.
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6. Member Lifecycle & GREENING ✓ (rules confirmed) / ? (values pending)

The greening flow (confirmed by owner 2026-09-09):

1. A visitor **signs up** under a sponsor (placement L/R) → account starts **RED (inactive)**.
2. They **shop** and place an order. The order (with **DP prices, a GST invoice, and the delivery address**) is **sent to WhatsApp** for the company. → SOON (WhatsApp send — to build, see §14)
3. The member **pays** (UPI / bank). When the **admin confirms payment received**, the order is processed.
4. **Greening rule:** an ID turns **GREEN only when the member's cumulative self-purchase reaches ≥ 25 SP** (activation_sp = 25). Example: buy 10 SP → still RED; the admin will **not** green the ID until total shopping reaches 25 SP.
5. **First purchase decides the capping** (max payout limit) for that ID — set at greening from the joining/first-purchase amount. See §9a. ? (exact SP→capping mapping pending business)

SP counts in the tree regardless of green status (confirmed): - The moment a member's order is confirmed, their **SP propagates up the binary tree** and contributes to every ancestor's leg — so **matching flows to the parent/upline** even if the buyer themselves is not yet green. A member can even build their own 100:100 downline.

Earning is gated on the earner's own green status (confirmed): - A member **personally receives Matching Bonus and Level Bonus only if their OWN ID is GREEN** (≥ 25 SP self-purchase). While RED, their own SP keeps accumulating (carry forward) but no matching/level payout is credited to them until they green.

Built today (● to be aligned to the above): - Green flag = is_active. Currently flips green at activation_sp SP (change 50 → 25) on admin order-confirm; sponsor gets ₹500 direct-referral on green. - **To implement:** (a) threshold 25 SP; (b) gate matching **and** level-bonus payout on earner-green; (c) capping set from first purchase.

? **Still open (for business):** 1. Exact **SP → capping ₹** mapping (25 SP → ₹?, 50 → ₹?, 100 → ₹?). Reference showed a ₹50,000 capping. 2. **Time window** to green after joining (does a RED ID ever lapse)? 3. **Monthly repurchase** to stay green/earning? SP + period + penalty if missed? 4. Joining **packages** (25/50/100 SP = ₹3,000/6,000/12,000) — is greening tied to buying a package, or any products totalling ≥25 SP?

7. Binary Tree & Placement

- Each member has a **sponsor** (who referred them → drives referral/level) and a **placement parent + position (L/R)**.
- New signup **spills over** down the chosen leg to the first open slot under the sponsor.
- **My Team** : left/right member counts, left/right **active** counts, parent, 3-level genealogy view.
- **?** Confirm placement rule: always spillover to outermost open slot on chosen leg? Or sponsor picks exact position? Any auto-balancing?

8. Selling Points (SP) / **?**

- Each product has an **SP** value. An order's `total_sp` = $\sum(\text{product.sp} \times \text{qty})$.
- On a **confirmed** order: buyer's `self_purchase_sp` increases; SP propagates up **every ancestor's** L or R leg (both `*_carry` for matching and lifetime `total*_sp` for rank).
- 1 SP $\approx$ ₹10 for matching payout (see §9). **?** Confirm SP→₹ for packages (100 SP pack = ₹12,000 in the plan sheet implies ₹120/SP business value — this is separate from the ₹10/SP matching rate).

9. Binary Matching & Weekly Payout (confirmed)

- **Rate: 1 SP = ₹10** (`matching_per_sp`). Simple calculation on matched SP.
- **Payout in ₹500 blocks**: minimum payout **₹500** and only in **multiples of ₹500** — i.e. per **50 matched SP** (50→₹500, 100→₹1,000, 150→₹1,500...). `matching_block_sp = 50` .
- **Carry forward**: unmatched SP and any sub-₹500 remainder **carries forward** to the next period. 
- **Weekly close** (admin "Run Weekly Close"): `matched = min(left_carry, right_carry)` , `closing = floor(matched/50)×50` , `payout = closing × ₹10` ; writes a **payout-register row** (Week, L SP, R SP, Matching SP, Closing SP, Payout, CF L, CF R) and credits the wallet.
- **Green gate (confirmed)**: a member is paid matching **only if their own ID is green** (§6). RED members' SP keeps carrying forward; it pays out once they green. *(to implement)*

9a. Capping principle / **?** values

- The member's **first purchase decides their capping** (a maximum payout limit for the ID), set at greening.
- Once capping is set, the ₹10/SP-in-₹500-blocks payout math applies within that cap.
- **?** Business to give the exact **first-purchase SP** → **capping ₹** table.

Other open: - **?** Should the weekly close run **automatically** on a fixed weekday (cron), not just manual?

10. Career Ranks & Rewards / **?**

- 12 ranks by cumulative SP on **both** legs (weaker leg governs):

Lvl	Rank	SP each leg	One-time bonus (₹)	Reward
1	Winner	200	4,000	Bag
2	Achiever	500	10,000	Smart Watch
3	Warrior	1,000	20,000	Tablet
4	Champion	2,000	40,000	Jim Corbett
5	Master	4,000	80,000	₹15k Mobile + Agra
6	Commander	8,000	80,000	₹30k Bike + Shimla
7	Royal Executive	16,000	2,50,000	₹1.5L Laptop + Gangtok
8	Imperial Leader	32,500	4,00,000	₹2L Gold + Goa
9	Diamond	65,000	6,00,000	₹3L Car + Bangkok
10	Crown	1,25,000	10,00,000	₹5L Car + Indonesia
11	King	2,50,000	15,00,000	₹7.5L Car + Switzerland
12	Global Icon	5,00,000	25,00,000	₹20L + Dubai

- Rank auto-detected for display; the **one-time cash bonus is paid manually by admin** ("Pay Rank Bonus"), mirroring the reference.
- ? Confirm: is rank purely from cumulative total SP each leg, or matched SP, or does admin approve rank? (Reference stored a `level` field — admin-set?)
- ? Are the "+₹15k Mobile / Bike / Car / Gold" amounts **cash funds** too, or physical gifts fulfilled offline? (Currently only the first number is cash; rest is recognition text.)
- ? Rank maintenance/repurchase to keep a rank?

11. Level Bonus & Direct Referral (present, green-gated) / ? (formula)

- Level Bonus is an active income** and, like matching, is **paid only when the earner's own ID is green** (§6). (to enable + green-gate)
- ? **Exact level-bonus definition needed from business:** is it the **career-rank achievement bonus** (§10 — the "Level Bonus" page in the reference), or a separate **generation/level income** (a %/amount on each level of downline SP/purchases)? Give the level→amount/percent table if the latter.
- Direct referral:** ₹500 to the sponsor when a directly-sponsored member greens (one-time). ? ₹500 per direct, only first, or per-leg (old sheet showed "Direct Ref 1/2 = ₹500")?
- ? Any **repurchase/retail** income, **pool/turnover** bonus, or **leadership override** beyond matching + level + rank + referral?

12. Direct-Selling (DSA) Portal

- Standalone agent (optional referral code, no tree). Greens on first purchase.
- Earns **40%** (`dsa_percent`) of the **taxable value (DP)** on their own confirmed orders → wallet.
- Dashboard: total sales, earnings, wallet, orders. Own orders + GST invoices.
- ? Confirm 40% flat, no downline/override for direct sellers.

13. Products & Catalog

- 28 products (SP/MRP/DP from your list) + 4 offer packages (₹5k/12.5k/25k/50k). Categories: Immunity, Digestive, Skin Care, Wellness, Personal Care, Home Care, Offers.
 - Admin CRUD (name, category, MRP, DP, cost, SP, stock, image, active, offer).
 - ? Offer-package SP values are **estimates** (40/100/200/400) — give real values.
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14. Orders & Payment (Admin-Confirm + WhatsApp) / SOON

- Member places order → **status** `pending` / `unpaid` . No SP/commissions yet.
 - **Order goes to WhatsApp** for the company with **DP prices, the GST invoice, and the delivery address** (confirmed flow).  SOON
to build — via a `wa.me` deep link and/or WhatsApp notification. ? company WhatsApp number + exact message format.
 - Member **pays via UPI/bank** (manual) per the Pay page. ? Should the member submit a **payment reference** for the admin to match?
 - Admin **Confirms Payment** → SP propagation, greening (if ≥ 25 SP), referral, DSA commission all run once. Or **Cancels** (only while unpaid).
 - No online gateway (owner's choice).
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15. GST Invoicing

- GST **18%, exclusive** (added on top of DP at checkout). GSTIN **09EJFPP4671A1Z5**, UP (state code 09), HSN default 30049011.
 - Tax invoice per order: CGST+SGST (intra-state) or IGST (inter-state), amount-in-words, printable.
 - Commissions computed on **taxable value**, not GST-inclusive total.
 - ? Per-product HSN/GST rate (currently one default rate + HSN for all)?
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16. KYC

- 5 upload slots: profile photo, Aadhaar front/back, PAN front/back. Stored in DB (≤ 5 MB, JPG/PNG/WEBP/PDF). Member uploads/replaces + previews; admin views. **No verification workflow** (per your instruction).
 - ? Should greening/withdrawal require KYC uploaded/verified?
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17. Wallet & Withdrawals / ?

- Earnings (matching + rank + referral + DSA) credit `wallet_balance` .
 - Member requests withdrawal (min ₹500 `payout_min`); admin approves → pays → rejects (refunds).
 - Settings exist for `admin_charge_percent` (5%) and `tds_percent` (5%) but ? **not yet applied** to withdrawals — confirm the deduction rules (TDS %, admin charge %, on what).
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18. Admin Panel

Overview (members, active, pending orders, confirmed revenue, pending payouts) · Orders (confirm/cancel) · Products (CRUD) · Members (block, rank bonus, KYC, reset password) · Payouts (weekly close + withdrawal requests) · MLM Settings (all rates) · Weekly Close. - ? Multiple admin roles/permissions needed, or single admin fine?

19. Configurable Settings (current defaults)

Key	Value	Meaning
matching_per_sp	10	₹ per matched SP
matching_block_sp	50	matching block size
daily_capping	0	matching cap (0=none)
direct_referral_bonus	500	₹ per direct referral
dsa_percent	40	direct-seller commission %
activation_sp	25	SP (cumulative self-purchase) to green an ID
repurchase_sp	25	(unused) monthly repurchase SP — ? rules
capping_by_first_purchase	?	first-purchase SP → capping ₹ table (pending)
gst_rate	18	GST %
price_gst_inclusive	false	DP is GST-exclusive
gstn	09EJFPP4671A1Z5	company GSTIN
payout_min	500	min withdrawal ₹
admin_charge_percent	5	(not yet applied)
tds_percent	5	(not yet applied)
level_bonus_percent	[]	level bonus — to be enabled , formula pending (§11)

Ranks live in code (`app/services/ranks.py`) — ? move to editable settings?

20. API Reference (current)

Auth: POST `/api/auth/signup` , `/auth/direct/signup` , `/auth/login` , `/auth/admin/login` , `/auth/token` **Member:** GET/PUT `/api/member/me` , `/member/dashboard` , `/member/direct/dashboard` , `/member/tree` , `/member/team` , `/member/level-bonus` , `/member/bonus` , `/member/payout-register` , POST `/member/change-password` , GET/POST `/member/kyc[/type][/file]` **Catalog:** GET `/api/catalog/products` , `/catalog/categories` , `/catalog/products/{slug}` **Orders:** GET/POST `/api/orders` , GET `/orders/{id}/invoice` **MLM:** GET `/api/mlm/commissions` , `/mlm/payouts` , POST `/mlm/payouts` , GET `/mlm/idcard` **Admin:** `/api/admin/overview` , `/members[...]` , `/members/{id}/{block,pay-rank-bonus,reset-password,kyc}` , `/orders[...]` , `/orders/{id}/{confirm,cancel}` , `/products[...]` , `/payouts` , `/payouts/close` , `/ranks` , `/settings` **Site:** GET `/api/site/settings` , GET `/api/health`

21. Status Summary

✓ Confirmed rules (to implement in the engine)

- Case-insensitive member-ID login.
- Greening = admin-confirmed payment **AND cumulative self-purchase ≥ 25 SP**.
- SP always counts in the tree & flows matching to the parent, regardless of buyer's green status.
- Member earns **matching + level bonus only when their OWN ID is green**.

- Matching = ₹10/SP, paid in ₹500 blocks (per 50 SP), remainder carries forward.
- First purchase decides the ID's capping.
- Level bonus is present and green-gated.
- Order sent to WhatsApp with DP + invoice + delivery address.

? Pending from business (kept as ? in the doc)

1. **Capping table** — first-purchase SP → capping ₹ (\$9a).
2. **Level-bonus definition/rates** — rank bonus or generation income? (\$11).
3. **Direct-referral** exact rule — per direct / first only / per-leg (\$11).
4. **Repurchase/maintenance** to stay green & keep rank (\$6, \$10).
5. **Greening time window** / ID lapse (\$6).
6. **TDS 5% + admin charge 5%** on withdrawals — how applied (\$17).
7. **Rank** — auto vs admin-approved; are +mobile/bike/car amounts cash or gifts (\$10).
8. **WhatsApp** number + message format (\$14).
9. **Offer-package SP** real values (\$13); **per-product HSN/GST** (\$15).
10. Auto weekly close (cron)? (\$9); any other income (pool/turnover/leadership)?
11.  SOON Offers / target-promotions section (deferred).

Confirmed items get built into the engine + tests; ? items go to the business team, then we finalize the backend.